MOBILE COMPUTING

ASSIGNMENT 1

C01

Design, Analyse and Validate an Efficient Cellular Network Strategy for a University Mobile Communication System

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Abstract

Mobile communication has become an essential part of modern educational institutions. Universities depend on cellular networks to provide reliable communication and internet connectivity for students, faculty members, administrative staff and visitors. These networks support various activities such as online learning, video conferencing, digital communication and access to educational platforms.

This assignment focuses on designing and analysing an efficient cellular network strategy for a university campus and its surrounding residential area. Three different cellular network approaches are considered: macrocell-based networks, small-cell networks and heterogeneous networks. Each approach is studied based on coverage, frequency reuse, interference management, network capacity, mobility and handoff operations.

The assignment also examines the requirements of supporting increasing numbers of mobile users, from 1,000 to 50,000 and above. Mathematical calculations are used to estimate traffic demand, network capacity, cell requirements and resource utilization. A Python-based analytical simulation is proposed to compare the performance of the three strategies under different user-load conditions.

Based on the analysis, a heterogeneous cellular network combining macro cells and small cells is selected as the most suitable approach. The final design is further modified to improve capacity, mobility management and service continuity during peak network traffic.

Keywords: Cellular Network, Mobile Communication, Macrocell, Small Cell, HetNet, Frequency Reuse, Handoff, 5G Network.

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TASK 1 – Design the Cellular Network Strategy (5 Marks)

1.1 Scenario and design objectives

The university network must support students, faculty, administrative staff and visitors as they move between teaching blocks, laboratories, libraries, hostels and nearby residential areas. It must carry voice, messaging and data traffic while maintaining acceptable coverage and service continuity. The design should scale from 1,000 users to 50,000 or more active users.

- Provide continuous outdoor coverage and dependable indoor coverage.
- Provide sufficient radio and backhaul capacity during busy hours.
- Control co-channel and adjacent-channel interference.
- Support mobility, handover and roaming without unnecessary interruption.
- Use spectrum and infrastructure efficiently and allow future expansion.

1.2 Strategy A – Macrocell-only network

A macrocell network uses a relatively small number of high-power base stations to cover large geographical areas. It is a coverage-oriented design and is useful for outdoor areas and lower-density locations.

Operating principle

16. A mobile station (MS) searches for a suitable cell and establishes radio access with the macro base station.
17. The base station provides the radio link and forwards traffic through the radio access network and transport network.
18. The mobile core performs subscriber, session, mobility and service functions.
19. When the user moves and a neighbouring cell becomes a better serving cell, the network prepares and executes a handover.

Cell and frequency planning

- For initial planning, assume a nominal macrocell radius of 1 km. Actual radius depends on terrain, antenna height, frequency band, clutter and link budget.

- Use a coordinated frequency plan. A reuse factor of N=1 may be used where the radio technology and interference coordination permit it; otherwise, reuse and sector planning must be selected from link-budget and interference analysis.
- Use antenna sectorisation, transmit-power control and coordinated scheduling to limit interference.
- Plan coverage overlap at cell boundaries so that handover can occur before the serving link becomes unusable.

Mobility and handover

The network uses measurement-based handover. The mobile station measures serving and neighbouring cell signal quality and reports measurements. The network selects a target cell, checks its available resources and commands the mobile station to move. In LTE and 5G, this is a network-controlled handover procedure rather than an informal disconnect-and-reconnect process.

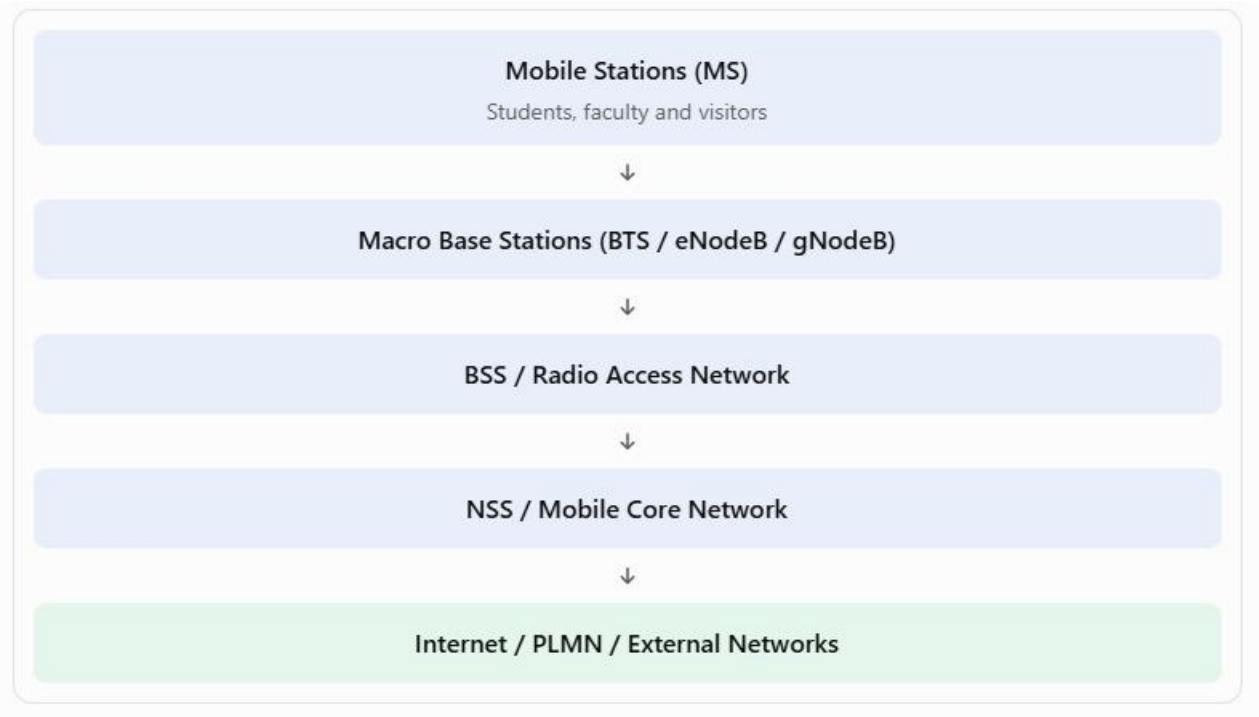


Figure 1. Simplified macrocell network architecture.

Components and responsibilities

Component	Role
Mobile Station (MS/UE)	User device; performs radio measurements and sends/receives data.
BTS / macro radio site	Provides radio coverage and radio transmission.
BSC (GSM) / RAN control functions	Controls radio resources in the relevant architecture.
MSC (GSM NSS)	Circuit-switched call control and mobility functions in GSM.
HLR/VLR (GSM)	Subscriber profile and visitor-location information.
LTE/5G core	Provides packet connectivity, authentication, mobility and session functions.

Advantages: wide coverage, fewer sites and relatively straightforward site management. Limitations: capacity can become inadequate in dense hotspots; users at cell edges may experience weaker signal; and large cells do not provide the same indoor capacity as a dense small-cell layer.

Assessment: this approach is suitable as a coverage foundation and may be adequate for a small population, but it is not sufficient as the only layer for the assumed 50,000-user peak demand.

1.3 Strategy B – Small-cell network

A small-cell network deploys many low-power radio nodes over smaller coverage areas. It is a capacity-oriented approach, particularly useful in libraries, lecture halls, laboratories, cafeterias and hostels.

Operating principle

20. Small cells are installed close to users in high-demand locations.

21. A device connects to a suitable small cell based on radio quality, access policy and available resources.
22. The small cell schedules radio resources and forwards traffic through backhaul to the mobile core.
23. As a user moves between small cells, the network performs handover to preserve the session.

Cell and frequency planning

- For the planning example, use a nominal small-cell radius of 0.2 km.
- Dense deployments can reuse the same carrier, but N=1 reuse requires effective interference coordination and power control.
- Coordinate neighbouring cells through radio-resource management; use ICIC/eICIC or the corresponding 5G coordination mechanisms where supported.
- Place cells according to measured traffic hotspots and indoor propagation, rather than relying only on geometric spacing.

Mobility and handover

Because cells are small, users may cross cell boundaries more often. Handover thresholds, hysteresis and time-to-trigger settings should prevent rapid back-and-forth handovers (ping-pong). The target cell should have enough capacity before the handover is commanded.

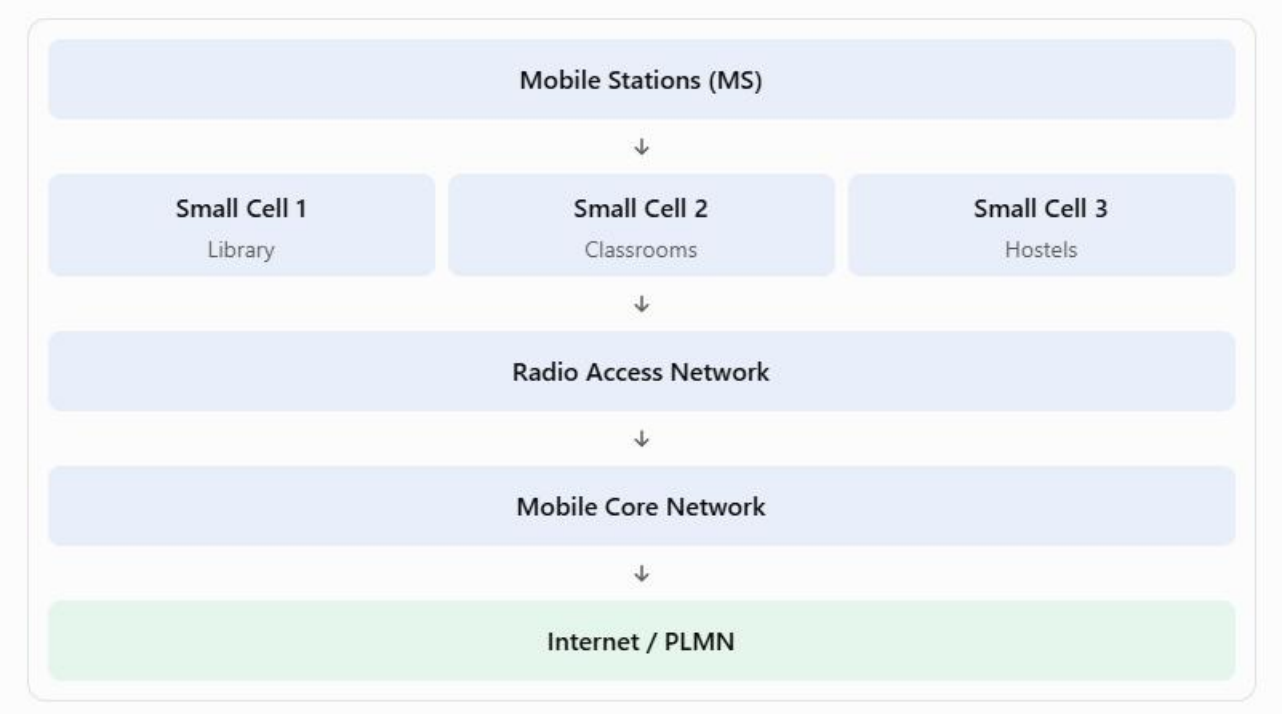


Figure 2. Simplified small-cell network architecture.

Advantages: high spatial reuse, improved indoor coverage and capacity close to users. Limitations: more sites, power and backhaul connections are needed; dense deployments can create interference; and frequent mobility can increase handover signalling.

Assessment: this approach is useful in high-density buildings but is less attractive as the sole coverage layer for a large campus and surrounding area.

1.4 Strategy C – Heterogeneous network (HetNet)

A heterogeneous network combines macro cells and small cells. Macro cells provide a broad coverage layer, while small cells add capacity where many users are concentrated. This approach separates the coverage role from the hotspot-capacity role.

Operating principle

24. Macro cells provide a coverage anchor across outdoor and lower-density areas.
25. Small cells are deployed in buildings and hotspots to serve concentrated traffic.
26. The radio network selects a suitable cell using signal quality, load, access policy and radio-resource availability.

27. Coordinated scheduling, power control and load balancing reduce congestion and manage interference.
28. Measurement-based handover moves users between cells while preserving service where possible.

Cell and frequency planning

- Use the planning radii of 1 km for macro cells and 0.2 km for small cells as initial assumptions only.
- Use coordinated spectrum allocation and reuse. A shared carrier with N=1 can be considered only where interference coordination and radio planning support it.
- Apply ICIC/eICIC or 5G inter-cell coordination, power control and load-aware scheduling.
- Use measured radio maps and busy-hour traffic maps to locate small cells.

Mobility and handover

Use measurement-based handover with hysteresis and time-to-trigger. Cell selection should consider both signal quality and target-cell load. A macro coverage layer can help maintain connectivity when a small-cell signal becomes weak, subject to the device, radio configuration and operator policy.

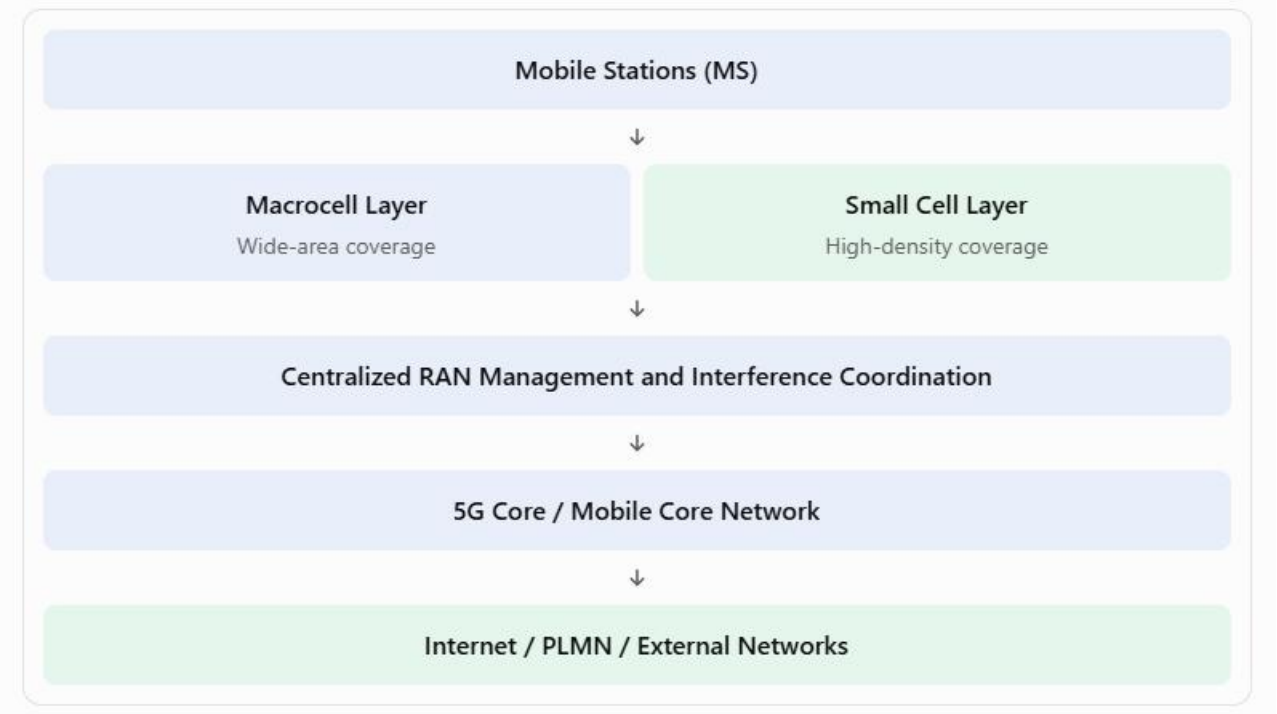


Figure 3. Simplified heterogeneous network architecture.

Component	Role in the HetNet
MS / UE	Measures cells, accesses the network and carries user traffic.
Macro gNodeB / eNodeB	Provides broad-area coverage and mobility support.
Small-cell gNodeB / eNodeB	Adds local capacity and indoor coverage.
RAN coordination	Coordinates radio resources, load balancing and handover.
AMF	Access and mobility management in 5G.
SMF	PDU session management.
UPF	User-plane packet forwarding.
UDM / AUSF	Subscriber data and authentication support.

Advantages: combines broad coverage with high hotspot capacity; supports scalable expansion; and can distribute traffic across layers. Limitations: higher design complexity, coordination requirements, backhaul needs and potentially higher initial cost.

Assessment: the HetNet is selected for further investigation because it offers the most balanced response to the campus coverage, capacity and mobility requirements.

1.5 Strategy comparison

Criterion	Macrocell	Small cell	HetNet
Coverage	Wide area	Local / indoor	Wide area plus hotspots
Capacity	Lower per dense area	High local capacity	High combined capacity
Handover activity	Generally lower	Generally higher	Depends on mobility and layer selection
Interference	Macro co-channel planning	Dense-cell coordination	Cross-layer coordination needed
Infrastructure	Fewer sites	Many sites	Macro layer plus small cells
Best use	Broad coverage	Crowded locations	Campus-wide mixed demand

TASK 2 – Identify the Basic Network Operation (2 Marks)

Strategy	Dominant operation/resource	Where it occurs	Why it matters as users grow
Macrocell	Radio-channel allocation and utilization	Between UE and macro radio site	More users share finite radio resources; congestion and interference increase.
Small cell	Handover and local	Between	Small coverage areas

	radio-resource allocation	neighbouring small cells and the RAN	increase cell-boundary crossings and signalling.
HetNet	Load-aware resource allocation and mobility coordination	Across macro/small-cell RAN and core mobility functions	Traffic can be shifted to available cells, but coordination and signalling must scale.

These operations are selected because they directly influence user throughput, delay, connection continuity and the amount of network capacity available during busy periods. As the population grows, radio scheduling becomes more competitive, access attempts become more frequent and mobility events create additional control-plane load.

TASK 3 – Analyse Network Capacity and Resource Requirements (6 Marks)

3.1 Planning assumptions

The question paper does not provide a measured traffic profile, site survey, spectrum licence or equipment specification. Therefore, the following values are explicit planning assumptions used to make a fair comparison. They should be replaced by measured values if the university has a real network survey.

Parameter	Assumption	Reason / note
Planning area	10 km²	Representative campus and nearby residential area.
Peak active users	50,000	Changed design requirement.
Average busy-hour data rate	0.25 Mbps/user	Simplified average across active users; not a per-user guaranteed rate.
Available spectrum	100 MHz	Illustrative spectrum budget; actual band and duplexing are not specified.

Macrocell nominal radius	1 km	Initial geometric planning assumption.
Small-cell nominal radius	0.2 km	Initial geometric planning assumption.
Usable capacity per macro site	1 Gbps	Illustrative aggregate planning value.
Usable capacity per small cell	0.4 Gbps	Illustrative aggregate planning value.
Capacity headroom target	30%	Planning margin for bursts and growth.

3.2 Offered traffic demand

Total offered traffic is estimated by multiplying the active-user count by the assumed average busy-hour rate.

$D = N \times R$

For 50,000 users: $D = 50{,}000 \times 0.25$ Mbps = 12,500 Mbps = 12.5 Gbps.

With a 30% headroom allowance: required planning capacity = $12.5 \times 1.30 = 16.25$ Gbps.

3.3 Macrocell capacity

Assume 12 macrocell sites, each with 1 Gbps usable aggregate capacity.

$C_{macro} = 12 \times 1$ Gbps = 12 Gbps.

Utilization = offered demand / capacity × 100 = 12.5 / 12 × 100 = 104.17%.

The estimated offered demand exceeds the assumed capacity. The macrocell-only design therefore cannot carry all the offered traffic at the target busy hour under these assumptions.

3.4 Small-cell capacity

Assume 40 small-cell sites, each with 0.4 Gbps usable aggregate capacity.

$C_{small} = 40 \times 0.4 = 16$ Gbps.

Utilization = 12.5 / 16 × 100 = 78.125%.

The small-cell design can carry the assumed demand, but its capacity is slightly below the 16.25 Gbps target that includes 30% headroom. It also needs careful mobility and interference planning.

3.5 HetNet capacity

Assume 12 macro sites and 30 small-cell sites.

Macro capacity = 12 × 1 = 12 Gbps. Small-cell capacity = 30 × 0.4 = 12 Gbps. Total = 24 Gbps.

Utilization = 12.5 / 24 × 100 = 52.08%. The HetNet exceeds the 16.25 Gbps planning target.

3.6 Geometric coverage estimate

For a circular cell, the idealized area is $A = \pi r^2$. A 1 km macro radius gives approximately 3.14 km² per cell. A 0.2 km small-cell radius gives approximately 0.1257 km² per cell.

Strategy	Sites	Ideal sum of circular areas
12 macrocells	12	12 × 3.14 = 37.68 km²
40 small cells	40	40 × 0.1257 ≈ 5.03 km²
12 macro + 30 small	42	37.68 + 3.77 ≈ 41.45 km²

These are sums of ideal circles, not unique covered land area. Real cells overlap, buildings attenuate signals, and terrain and antenna patterns alter coverage. A radio survey and propagation model are required to determine actual site count and placement.

3.7 Frequency reuse and interference

A reuse factor of N=1 means the same carrier can be reused in every cell. It improves spectrum reuse but may increase co-channel interference, especially in dense small-cell deployments. The network should use sectorisation where appropriate, coordinated scheduling, transmit-power control, inter-cell interference coordination and careful channel planning. Adjacent-channel leakage and receiver selectivity should also be considered. The exact reuse plan cannot be finalized without the band, bandwidth, radio technology, antenna pattern and link-budget details.

3.8 Random-access channel demand

For an illustrative access-load estimate, assume 5% of 50,000 active users initiate a new access attempt in a one-second interval. That gives 50,000 × 0.05 = 2,500 attempts per second across the network. If spread evenly across 42 HetNet sites, the mean is about 59.5 attempts per site per second. Actual PRACH capacity depends on the NR configuration, number of preambles, time/frequency occasions, contention and retransmission rate. This estimate is an offered-load indicator, not a claim that every cell needs exactly 60 preambles each second.

3.9 Handover activity

For comparison, assume average handovers per user per hour of 0.5 for macrocell, 3 for small cell and 1.5 for HetNet. At 50,000 users:

Strategy	Assumed handovers/user/hour	Estimated handovers/hour
Macrocell	0.5	25,000
Small cell	3.0	150,000
HetNet	1.5	75,000

These are assumed mobility rates for comparative modelling. Real rates depend on user speed, cell geometry, route, handover thresholds and time spent indoors or outdoors.

3.10 Planning and signalling overhead

The number of sites is a simple indicator of configuration effort: 12 macro sites, 40 small cells or 42 combined sites in the original HetNet. This does not capture all operational effort. Each site needs radio planning, neighbour relations, backhaul, power, monitoring, security, software configuration and acceptance testing. Signalling overhead is driven by registration, measurement reports, handover preparation, access retries and session management. It should be measured in a detailed simulator or field trial rather than assigned an unsupported fixed percentage.

Measure	Macrocell	Small cell	Original HetNet
Site count	12	40	42
Aggregate assumed capacity	12 Gbps	16 Gbps	24 Gbps
Demand at 50,000 users	12.5 Gbps	12.5 Gbps	12.5 Gbps
Utilization	104.17%	78.13%	52.08%
30% headroom target met?	No	No	Yes
Handover events/hour (assumed)	25,000	150,000	75,000

Task 3 conclusion: under the declared assumptions, the HetNet has the strongest capacity position and meets the planning headroom target. The small-cell-only approach carries the nominal demand but falls slightly short of the 30% headroom target. The macrocell-only approach is overloaded.

TASK 4 – Best, Average and Worst-Case Operating Conditions (4 Marks)

Strategy	Best case	Average case	Worst case
Macrocell	Few users, good signal, low interference and few handovers.	Moderate user density and ordinary movement; service remains usable.	Peak demand overloads radio resources; edge users and co-channel interference reduce quality.
Small cell	Users are well distributed and remain within cells; strong indoor signal.	Normal movement between buildings creates regular handovers but cells	Crowded cells, rapid movement, access contention and interference increase handover failures and

		retain capacity.	delay.
HetNet	Macro layer covers the area and small cells absorb hotspot traffic; coordination is effective.	Users move normally and load is shared between layers.	Several hotspots overload simultaneously or coordination/backhaul is inadequate; handover and interference still cause degradation.

Best-case conditions occur when offered traffic is below capacity, radio conditions are favourable and users are distributed as expected. Average conditions include ordinary busy-hour traffic and normal campus movement. Worst-case conditions occur during events, examinations, class changes or emergencies when many users request service in the same area at the same time.

Interference affects all three designs, but dense small-cell deployments are especially sensitive to poor coordination. Access-channel contention increases when many devices attempt to connect together. Handover frequency rises as cell size decreases. Roaming between different public land mobile networks (PLMNs) can add authentication, policy and routing steps; it is distinct from ordinary handover between cells in the same network.

TASK 5 – Comparative Performance Analysis (3 Marks)

The network is compared at 1,000, 10,000, 25,000 and 50,000 users. The average traffic assumption is 0.25 Mbps per user, so the offered demand rises from 0.25 Gbps to 12.5 Gbps.

Performance criterion	Macrocell	Small cell	HetNet
Throughput	Limited by the 12 Gbps aggregate capacity in the model.	Limited by 16 Gbps aggregate capacity.	Limited by 24 Gbps aggregate capacity.
Delay	Rises sharply as	Lower at moderate	Lower modelled delay

	capacity is approached or exceeded.	load, but handover and contention can add delay.	because the assumed capacity headroom is larger.
Connection reliability	Risk of congestion at peak load.	Risk from overloaded individual cells and frequent handovers.	Better load distribution, provided target cells and backhaul have capacity.
Handover	Fewer cell crossings due to larger cells.	More frequent due to smaller cells.	Intermediate in the assumed model.
Interference	Macro co-channel planning is important.	Dense-cell interference needs careful coordination.	Requires coordination between macro and small-cell layers.
Scalability	Adding capacity may require additional macro sites or spectrum.	Capacity can be added locally, but site density and handovers grow.	Can expand by adding small cells where demand is measured.

At 1,000 users, all three designs have ample capacity under the assumptions. At 10,000 users, the demand is 2.5 Gbps and each strategy remains below its aggregate capacity. At 25,000 users, demand is 6.25 Gbps and all three remain within capacity. At 50,000 users, demand reaches 12.5 Gbps: the macrocell model is overloaded, the small-cell model is at 78.13% utilization, and the HetNet is at 52.08%.

For the university, throughput alone is not enough. The network must also keep online examinations, voice calls and learning applications stable while users move. The HetNet is preferred because it combines coverage with capacity and can direct hotspot traffic to small cells.

TASK 6 – Formulate the Mobility and Handoff Procedure (3 Marks)

6.1 Handoff roles

Entity	Role
Serving cell / source gNodeB	Currently serves the UE and evaluates its measurement report.
Target cell / target gNodeB	Accepts or rejects the handover based on radio and admission resources.
UE	Measures serving and neighbouring cells; follows the handover command.
AMF / core	Supports mobility and control-plane procedures when required by the handover type.
UPF	May have its user-plane path updated as part of path switching.

6.2 Procedure

29. The UE is connected to a serving macro or small cell and has an active session.
30. The UE measures serving and neighbouring-cell signal quality according to configured measurement rules.
31. When the configured event condition, hysteresis and time-to-trigger are satisfied, the UE sends a measurement report.
32. The serving gNodeB evaluates the report and selects a target cell using radio quality, load and admission policy.
33. The source gNodeB sends a handover request to the target gNodeB over the applicable inter-node interface.
34. The target checks admission control and reserves the required radio resources. It returns an acknowledgement if it can accept the UE.
35. The source sends the handover command to the UE.

36. The UE synchronizes with and accesses the target cell, then completes the handover.
37. The network performs any required path switch and releases the source-cell resources.

6.3 Handover trigger

An illustrative trigger is that target-cell RSRP exceeds serving-cell RSRP by a configured margin, for example 3 dB, continuously for a configured time-to-trigger. This is an example parameter only; real thresholds must be tuned to the radio environment and mobility profile. RSRP, RSRQ and SINR may all be relevant.

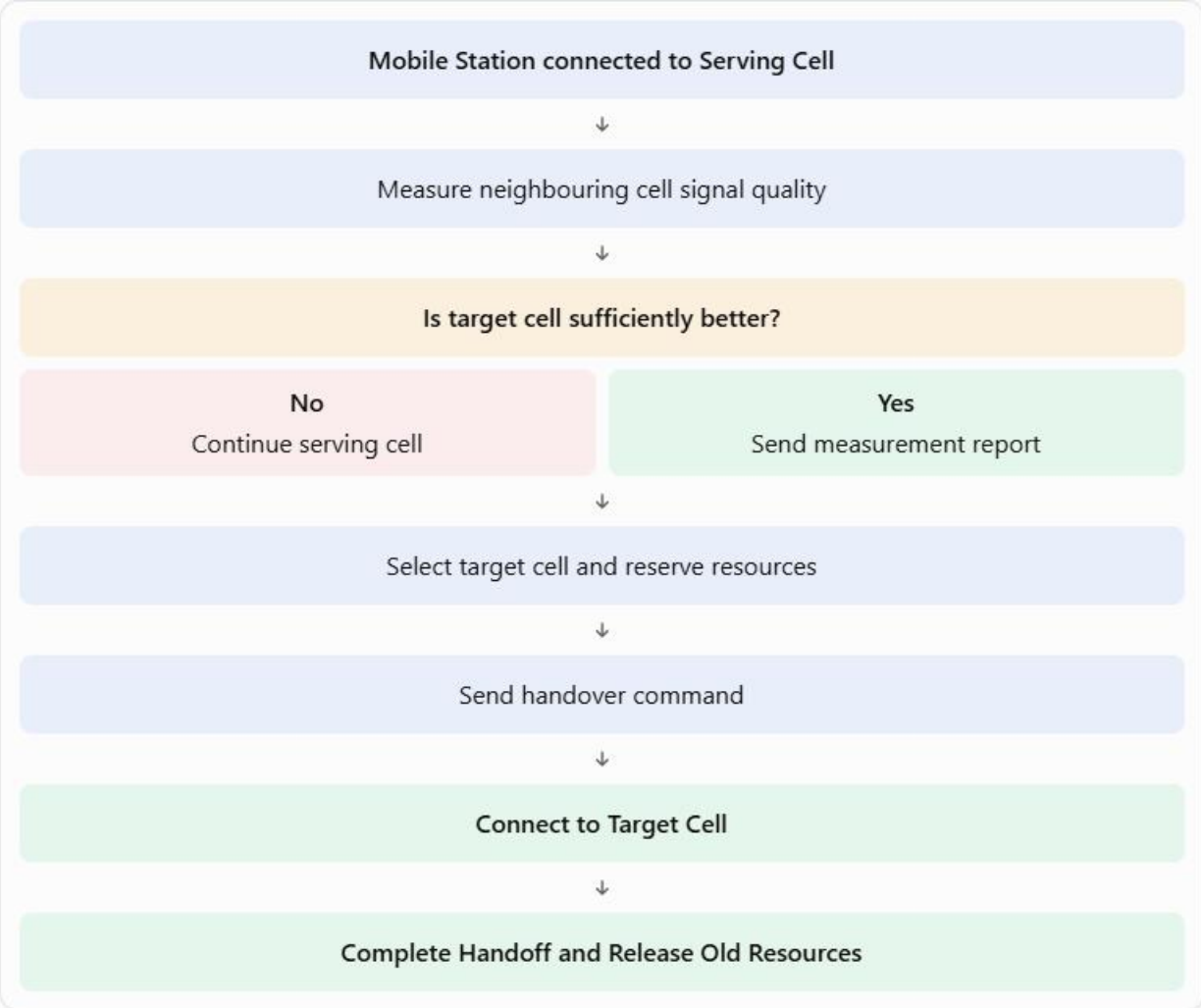


Figure 4. Measurement-based handover sequence.

6.4 Required resources and failure cases

- Radio resources and admission capacity in the target cell.
- Control-plane signalling between source, target and core where applicable.
- Reliable transport/backhaul and user-plane path switching.
- Failure causes include target-cell congestion, weak target signal, signalling timeout, radio-link failure and excessive interference.
- The design reduces interruption by preparing the target before the UE leaves the serving cell and by avoiding unnecessary handovers.

Handover is not the same as roaming. Handover changes the serving cell or access node within a network or mobility domain. Roaming generally involves a subscriber using a visited PLMN and may require inter-operator authentication, policy and charging arrangements.

TASK 7 – Protocol Stack Architecture and Performance Impact (4 Marks)

The selected design uses a 5G NR radio access network and 5G Core as the reference architecture. A HetNet can include macro and small-cell radio nodes connected to the same core. The exact protocol stack depends on the deployed 3GPP release and network configuration.

7.1 User-plane protocol layers

Layer	Main function
Application	Provides learning platforms, video, messaging and other services.
IP	Provides packet addressing and routing.
SDAP	Maps QoS flows to data radio bearers.
PDCP	Provides functions such as ciphering, integrity protection where applicable, sequence handling and header compression.
RLC	Segments and reassembles data; supports retransmission modes where configured.

MAC	Multiplexes logical channels, performs scheduling-related functions and manages radio access.
PHY	Transmits and receives radio signals over the air interface.

7.2 Control-plane protocols

Protocol	Main function
RRC	Controls radio connection, measurement configuration, mobility and radio-bearer configuration.
NAS	Supports UE-to-core procedures such as registration, authentication-related signalling and mobility/session management.
NGAP	Carries control signalling between NG-RAN and the 5G Core over N2.
XnAP	Supports control-plane signalling between NG-RAN nodes over Xn where deployed.



Figure 5. Simplified 5G user-plane flow.

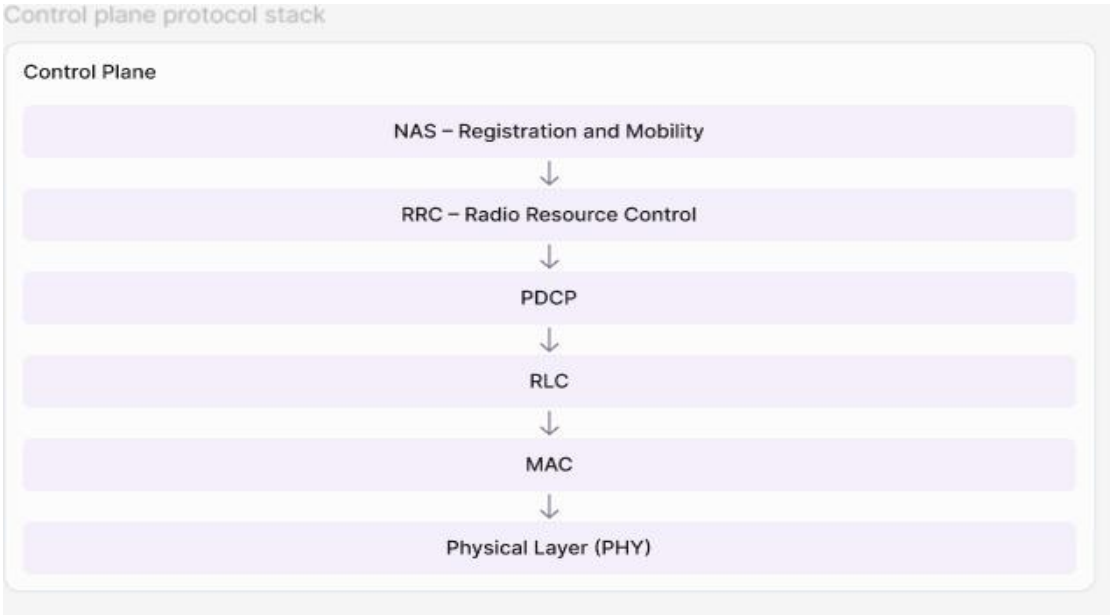


Figure 6. Simplified 5G control-plane flow.

7.3 Control plane and user plane

The control plane establishes and manages connectivity. It handles registration, authentication, radio connection control, mobility and session control. The user plane carries application packets such as video, web traffic and learning-platform data. In 5G, the AMF is central to access and mobility management, the SMF manages sessions, and the UPF forwards user traffic.

7.4 Handover and protocol interaction

The UE reports radio measurements through RRC. The source gNodeB decides whether a handover is needed and coordinates with the target gNodeB using Xn signalling where available, or through core-assisted procedures involving N2. Once the target is prepared, the UE receives a handover command and accesses the target. The user-plane path is then updated as required.

7.5 Performance and interoperability considerations

- PHY and radio conditions influence coverage, spectral efficiency and achievable throughput.
- MAC scheduling affects queueing delay and fairness between users.
- RLC retransmissions can improve reliability but add delay under poor radio conditions.
- RRC measurement and handover procedures create signalling overhead; poor parameter tuning can cause ping-pong handovers.
- Core and transport latency affect end-to-end service performance.
- Interoperability requires compatible radio features, software releases, interfaces, security settings and configuration across vendors.

The protocol stack therefore supports both data transfer and network control. Efficient coordination is essential for low delay and service continuity, particularly when users move between macro and small cells.

TASK 8 – Model and Test the Proposed Strategies (4 Marks)

A reproducible analytical simulation is used to compare the three strategies under equivalent assumptions. It calculates offered demand, throughput limited by aggregate capacity, capacity utilization, an illustrative delay estimate and handover-event counts. It is a planning model, not a packet-level 5G simulator.

8.1 Software and test setup

Item	Specification
Language	Python 3
Libraries	NumPy, Pandas and Matplotlib
User loads	1,000; 10,000; 25,000; 50,000
Average offered rate	0.25 Mbps per user
Capacity assumptions	12 Gbps macro; 16 Gbps small cell; 24 Gbps HetNet
Mobility assumptions	0.5; 3.0; 1.5 handovers/user/hour respectively

8.2 Test cases

Test	User count	Traffic demand	Purpose
TC1	1,000	0.25 Gbps	Low-load baseline
TC2	10,000	2.50 Gbps	Moderate-load comparison
TC3	25,000	6.25 Gbps	High-load comparison
TC4	50,000	12.50 Gbps	Peak-load design challenge

Normal mobility is represented by the assumed average handover rates. A high-mobility stress test should increase those rates and compare the resulting signalling load; this simplified model reports the assumed handover events but does not simulate individual handover success or

interruption. Frequency reuse and interference are represented only by an illustrative index, not by a physical propagation calculation.

8.3 Execution steps

38. Install Python 3 and the required libraries using the command shown in Appendix A.
39. Save the supplied code as cellular_simulation.py.
40. Run python cellular_simulation.py from the folder containing the file.
41. Review the printed table and the generated throughput, delay and utilization plots.
42. Repeat with altered mobility rates or capacity values to perform sensitivity analysis.

8.4 Model limitations

The model assumes aggregate capacity per site and an average user data rate. It does not model individual radio resource blocks, fading, building penetration, MIMO layers, scheduler details, PRACH collisions, packet retransmissions, QoS classes, backhaul bottlenecks or actual handover signalling. Therefore, its results support comparative reasoning but cannot replace a detailed system-level simulator or field measurements.

TASK 9 – Record Experimental Performance (3 Marks)

The following tables show the outputs calculated by the Task 8 analytical model. They are labelled as simulated/modelled values to distinguish them from field measurements.

9.1 Throughput and utilization

Users	Macrocell (Gbps)	Small cell (Gbps)	HetNet (Gbps)
1000	0.25	0.25	0.25
10000	2.50	2.50	2.50
25000	6.25	6.25	6.25
50000	12.00	12.50	12.50

Users	Macrocell	Small cell	HetNet
1000	2.08%	1.56%	1.04%
10000	20.83%	15.62%	10.42%
25000	52.08%	39.06%	26.04%
50000	104.17%	78.12%	52.08%

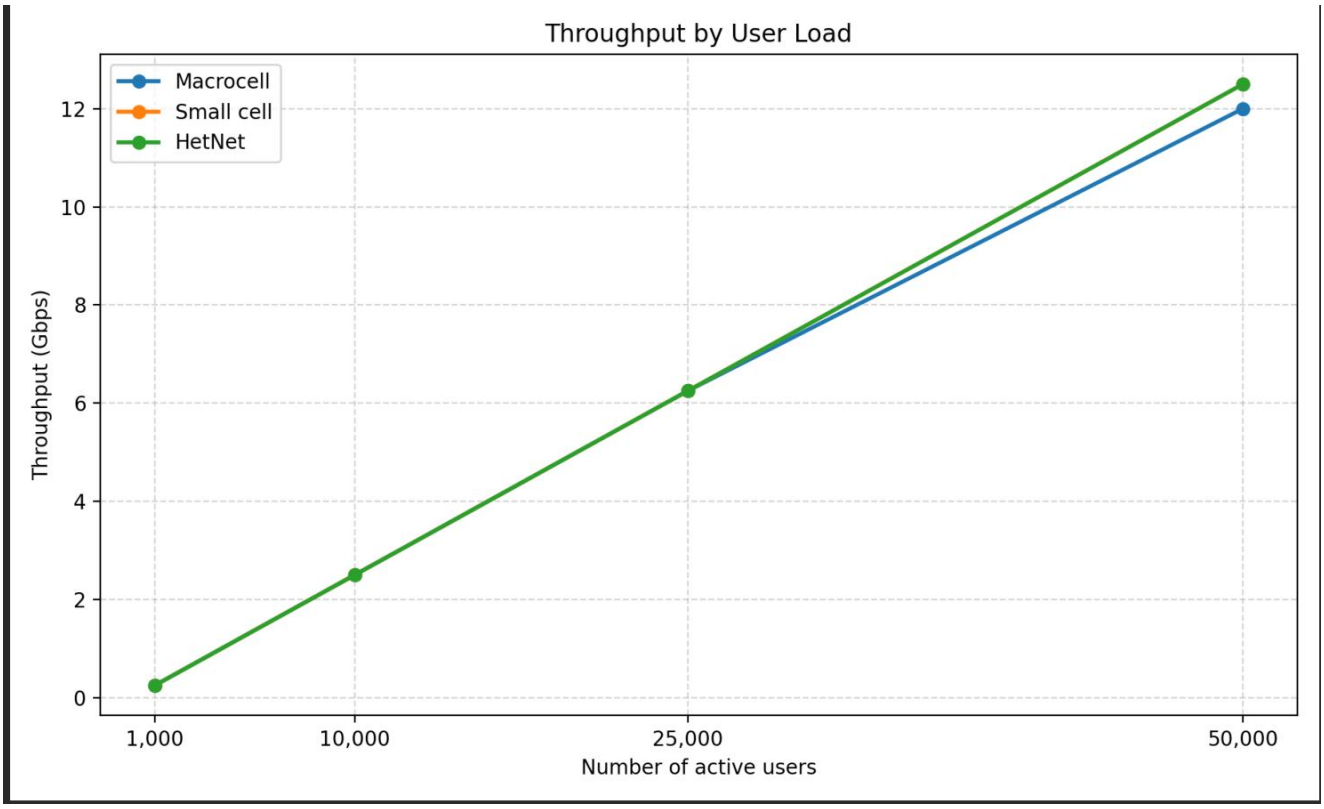


Figure 7: Throughput Comparison of Different Cellular Network Strategies

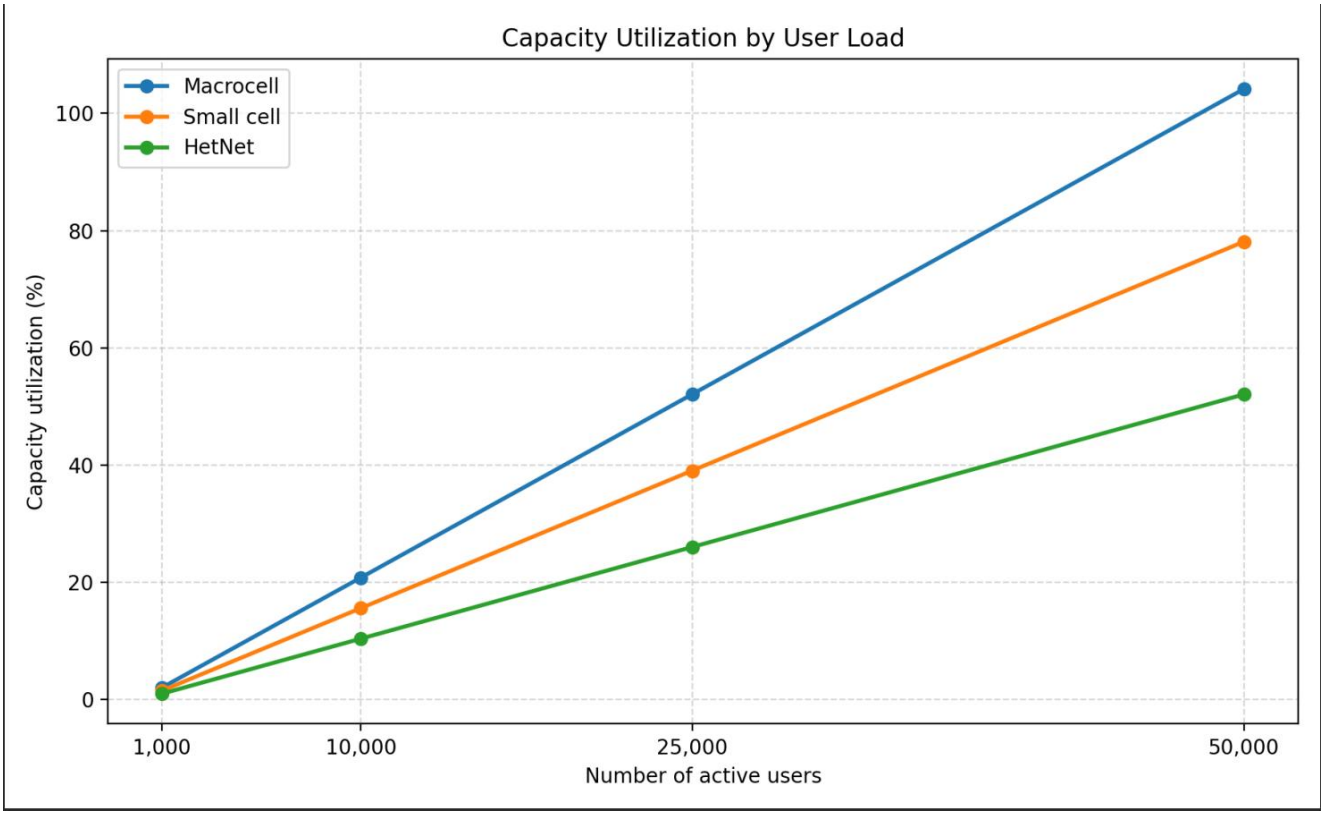


Figure 8: Network Resource Utilization Comparison

9.2 Illustrative delay estimate

Users	Macrocell (ms)	Small cell (ms)	HetNet (ms)
1000	20.0	20.0	20.0
10000	22.7	21.4	20.6
25000	48.3	32.5	24.6
50000	200.0	159.5	48.3

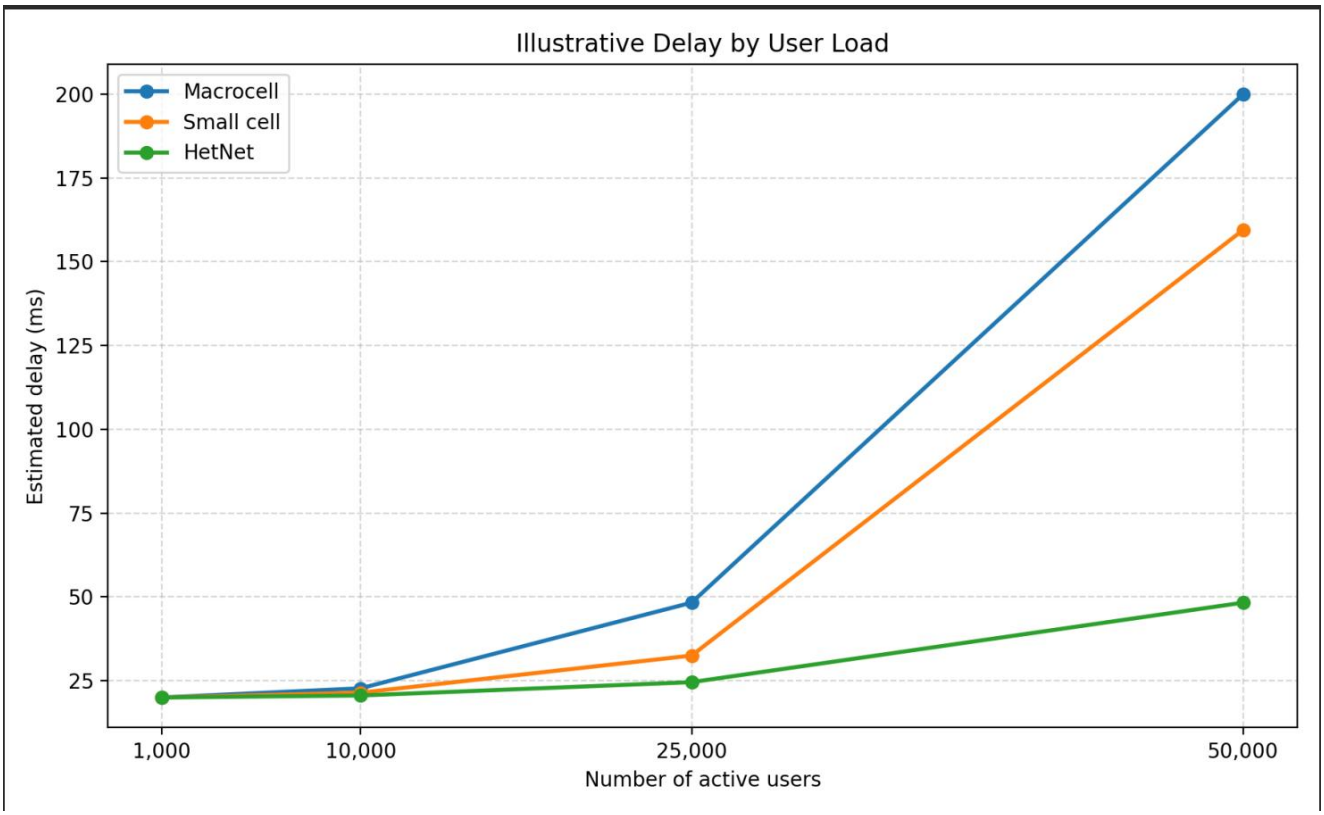


Figure 9: Network Delay Comparison Under Different User Loads

9.3 Handoff activity at 50,000 users

Strategy	Assumed handovers/user/hour	Estimated events/hour
Macrocell	0.5	25,000
Small cell	3.0	150,000
HetNet	1.5	75,000

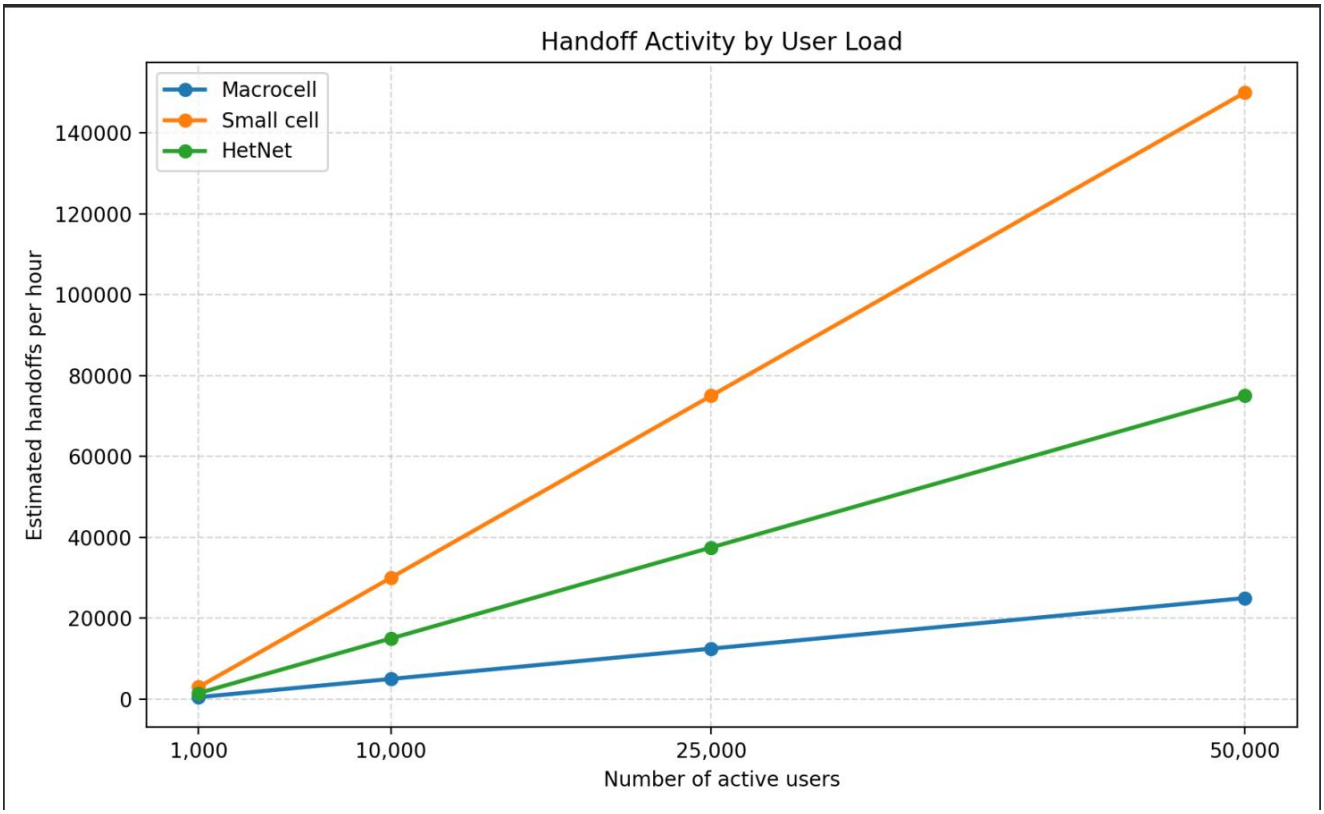


Figure 10: Handoff Rate Comparison of Different Cellular Network Strategies

The delay formula used in the code is a simple queueing-style illustration: $T = 20 + 50U^2/(1-U)$, where U is utilization, with a 200 ms cap applied at $U \geq 0.95$. It is not a validated 5G latency model. The throughput is min(offered demand, aggregate capacity). The handover figures are calculated from assumed mobility rates. A real call-drop rate, PRACH collision rate, SINR or handover interruption time cannot be claimed from this model.

TASK 10 – Compare Expected and Observed Behaviour (4 Marks)

Strategy	Expected from capacity analysis	Observed in the model	Interpretation
Macrocell	Good coverage but may lack peak capacity.	At 50,000 users, 12.5 Gbps demand exceeds 12 Gbps capacity.	The model caps throughput at 12 Gbps and indicates overload.
Small cell	Higher local capacity	Carries 12.5 Gbps	Capacity is adequate

	with more mobility events.	demand at 78.13% utilization; 150,000 assumed handovers/hour.	for demand, but handover signalling is the main concern.
HetNet	Balanced coverage and capacity.	Carries demand at 52.08% utilization; 75,000 assumed handovers/hour.	The largest capacity margin is available in this comparison.

The broad observed pattern agrees with the original expectations. The macrocell design becomes capacity-limited; the small-cell design carries the offered traffic but has the highest assumed handover rate; and the HetNet provides the strongest balance. Any difference between a real deployment and these results could arise from uneven traffic distribution, interference, backhaul limits, access contention, scheduling, building penetration and signalling load.

The model should not be interpreted as proving a specific real-world call-drop percentage or end-to-end delay. Those measures require more detailed modelling and/or field testing.

TASK 11 – Recommend and Defend the Final Design (4 Marks)

The recommended design is the heterogeneous network consisting of a macro coverage layer and small cells in high-demand locations.

11.1 Technical justification

- The original HetNet has an assumed aggregate capacity of 24 Gbps, exceeding the 12.5 Gbps offered demand.
- Its calculated utilization is 52.08%, lower than the small-cell-only design at 78.13%.
- The macro layer provides broad-area coverage while small cells add capacity in indoor and crowded locations.
- Load-aware cell selection and coordinated radio scheduling can distribute traffic and reduce localized congestion.

- Measurement-based handover can support movement between cells, provided target-cell resources and backhaul are available.

11.2 Infrastructure

Infrastructure element	Planning requirement
Macro radio sites	12 sites in the initial design.
Small cells	30 sites in the initial design, located using traffic and coverage surveys.
Spectrum	100 MHz illustrative planning budget; actual licensing and band plan must be verified.
Backhaul	Fibre or suitable high-capacity wireless links with sufficient aggregate capacity.
Core	5G Core functions for access, mobility, session management and user-plane forwarding.
Operations	Monitoring, power, backup, security, software lifecycle and fault management.

The HetNet has greater coordination complexity and site count than the macro-only design. However, it offers a more practical balance for a university with both broad outdoor areas and concentrated indoor demand. The recommendation is conditional on a detailed radio survey, licensed-spectrum confirmation, backhaul design and vendor-specific capacity validation.

TASK 12 – Re-evaluate and Adapt the Design (8 Marks)

A. Impact analysis (2 Marks)

The changed requirement specifies 50,000 active users during peak hours across the campus and nearby residential area. Compared with a small initial deployment, this increases aggregate traffic demand, the number of simultaneous access attempts and the number of users moving between cells. The 100 MHz assumed spectrum budget remains limited, so capacity cannot be increased simply by allocating unlimited new frequencies.

- Traffic demand rises to an estimated 12.5 Gbps using 0.25 Mbps per active user.
- A 30% planning margin raises the target aggregate capacity to 16.25 Gbps.
- More users in libraries, hostels and lecture halls create localized hotspots even if average campus-wide load is moderate.
- Frequent movement increases handover signalling and target-cell admission pressure.
- Frequency reuse improves spectrum efficiency but can increase co-channel interference unless coordinated.

B. Evaluation of the original design (2 Marks)

The original recommended HetNet contains 12 macro sites and 30 small cells, with an assumed aggregate capacity of 24 Gbps. This exceeds the 16.25 Gbps planning target and can carry the 12.5 Gbps offered demand in the aggregate model. Its utilization at peak demand is 52.08%.

However, aggregate capacity does not guarantee that every location has enough capacity. A library or hostel can become congested while unused capacity remains elsewhere. The original design therefore remains a sound foundation but should be adjusted to address hotspot distribution, handover activity and interference. The macro layer should remain for continuity, while additional small cells should be placed where traffic measurements justify them.

C. Adaptation or redesign (2 Marks)

Decision: MODIFY the original HetNet rather than replace it. Increase the small-cell layer from 30 to 40 sites while retaining 12 macro sites. The ten additional small cells should be placed in

measured hotspots such as high-density hostels, libraries, laboratories, lecture halls and nearby residential clusters.

Element	Original HetNet	Modified HetNet
Macro sites	12	12
Small-cell sites	30	40
Total sites	42	52
Macro capacity	12 Gbps	12 Gbps
Small-cell capacity	12 Gbps	16 Gbps
Total assumed capacity	24 Gbps	28 Gbps

The revised architecture also uses load-aware cell selection, coordinated scheduling, power control, hysteresis and time-to-trigger tuning, and monitoring of backhaul and target-cell admission. The same 100 MHz assumed spectrum budget is reused more carefully; no additional spectrum is assumed.

D. Evidence-based defence (2 Marks)

Revised aggregate capacity = (12 macro sites × 1 Gbps) + (40 small cells × 0.4 Gbps) = 12 + 16 = 28 Gbps.

Revised utilization = 12.5 / 28 × 100 = 44.64%.

Revised capacity headroom = 28 − 12.5 = 15.5 Gbps. The 30% headroom target is 16.25 Gbps, so the revised capacity exceeds that target by 11.75 Gbps.

Measure	Original HetNet	Modified HetNet
Macro sites	12	12
Small cells	30	40
Total assumed capacity	24 Gbps	28 Gbps
Offered demand	12.5 Gbps	12.5 Gbps
Utilization	52.08%	44.64%
Capacity above demand	11.5 Gbps	15.5 Gbps

30% headroom target met?	Yes	Yes

The modification improves aggregate capacity and provides more local capacity where hotspots are identified. It does not automatically guarantee lower handover rates or interference: those depend on cell placement, mobility patterns, radio configuration and coordination. The final decision is therefore to modify the HetNet, validate the new locations with a radio survey, and then confirm performance through a detailed system-level simulation and field acceptance tests.

Overall Conclusion

This assignment compared macrocell-only, small-cell-only and heterogeneous cellular network strategies for a university mobile communication system. The analysis considered coverage, frequency reuse, interference, capacity, access-channel load, mobility, handover and 5G protocol functions. Under the stated assumptions, 50,000 active users generate 12.5 Gbps of busy-hour traffic, with a 16.25 Gbps capacity target when a 30% margin is included. The macrocell-only option has insufficient assumed capacity. The small-cell-only option can carry the nominal demand but has a higher assumed handover rate and falls slightly short of the 30% capacity-margin target. The original HetNet has 24 Gbps capacity and meets the target. The final design modifies it by increasing the small-cell count from 30 to 40, resulting in an assumed capacity of 28 Gbps. This is the preferred design because it combines broad coverage with additional capacity in crowded locations. Actual deployment must be validated with licensed-spectrum information, site surveys, detailed radio planning, backhaul design and field measurements.

References

43. Rappaport, T. S., Wireless Communications: Principles and Practice, 2nd ed., Prentice Hall, 2002.
44. Schiller, J. H., Mobile Communications, 2nd ed., Pearson Education, 2003.
45. 3GPP TS 38.300, NR and NG-RAN Overall Description; consult the applicable release.
46. 3GPP TS 38.331, NR Radio Resource Control (RRC) Protocol Specification; consult the applicable release.
47. 3GPP TS 23.501, System Architecture for the 5G System; consult the applicable release.

48. 3GPP TS 38.401, NG-RAN Architecture Description; consult the applicable release.

Appendix A – Complete Python Simulation Code

Install dependencies in a terminal:

```
python -m pip install numpy pandas matplotlib
```

Save the following as cellular_simulation.py and run it with python cellular_simulation.py.

CODE:

task09_implementation.py

C: > Users > Sravanthi > Downloads > task09_implementation.py

```python
"""Task 9: Cellular Network Strategy - performance implementation.

This reproduces the Task 9 modelled results for Macrocell, Small-cell and
HetNet strategies. Values are illustrative assumptions, not field measurements.
Run: python task09_implementation.py
"""

from pathlib import Path
import pandas as pd
import matplotlib.pyplot as plt

# ---------------------- 1. INPUT PARAMETERS ----------------------
USER_LOADS = [1_000, 10_000, 25_000, 50_000]
AVERAGE_RATE_MBPS = 0.25

STRATEGIES = {
    "Macrocell": {"capacity_gbps": 12.0, "handovers_per_user_hour": 0.5},
    "Small cell": {"capacity_gbps": 16.0, "handovers_per_user_hour": 3.0},
    "HetNet": {"capacity_gbps": 24.0, "handovers_per_user_hour": 1.5},
}

OUTPUT_DIR = Path(__file__).resolve().parent / "task09_outputs"
OUTPUT_DIR.mkdir(exist_ok=True)

# ---------------------- 2. CALCULATE RESULTS ----------------------
rows = []
for users in USER_LOADS:
    demand_gbps = users * AVERAGE_RATE_MBPS / 1000

    for strategy, config in STRATEGIES.items():
        capacity = config["capacity_gbps"]
        utilization = demand_gbps / capacity
        throughput = min(demand_gbps, capacity)

        # Illustrative delay formula used in the assignment model.
        # It is NOT a validated 5G latency formula.
```

```
    for strategy, config in STRATEGIES.items():
        if utilization >= 0.95:
            delay_ms = 200.0
        else:
            delay_ms = 20 + (50 * utilization**2 / (1 - utilization))

        handovers = round(users * config["handovers_per_user_hour"])

        rows.append({
            "Users": users,
            "Strategy": strategy,
            "Demand_Gbps": demand_gbps,
            "Throughput_Gbps": throughput,
            "Utilization_Percent": utilization * 100,
            "Estimated_Delay_ms": delay_ms,
            "Handoffs_Per_Hour": handovers,
        })

df = pd.DataFrame(rows)

# Save complete experiment data.
df.to_csv(OUTPUT_DIR / "task09_all_results.csv", index=False)

# Create wide-format tables similar to the Task 9 tables in the report.
for metric, filename in [
    ("Throughput_Gbps", "throughput_table.csv"),
    ("Utilization_Percent", "utilization_table.csv"),
    ("Estimated_Delay_ms", "delay_table.csv"),
]:
    table = df.pivot(index="Users", columns="Strategy", values=metric)
    table.to_csv(OUTPUT_DIR / filename)

handoff_table = pd.DataFrame([
    {
        "Strategy": name,
```

```
        {
            "Assumed handovers/user/hour": cfg["handovers_per_user_hour"],
            "Estimated events/hour at 50,000 users": round(
                50_000 * cfg["handovers_per_user_hour"]
            ),
        }
        for name, cfg in STRATEGIES.items()
    ])
    handoff_table.to_csv(OUTPUT_DIR / "handoff_table.csv", index=False)

    # ---------------------- 3. PRINT TABLES ----------------------
    print("\nTASK 9 - CELLULAR NETWORK PERFORMANCE RESULTS")
    print("All results below are modelled estimates based on stated assumptions.\n")
    print("COMPLETE RESULTS")
    print(df.round(2).to_string(index=False))

    for metric, heading in [
        ("Throughput_Gbps", "THROUGHPUT (Gbps)"),
        ("Utilization_Percent", "CAPACITY UTILIZATION (%)"),
        ("Estimated_Delay_ms", "ILLUSTRATIVE DELAY (ms)"),
    ]:
        print(f"\n{heading}")
        print(df.pivot(index="Users", columns="Strategy", values=metric).round(2))

    print("\nHANDOFF ACTIVITY AT 50,000 USERS")
    print(handoff_table.to_string(index=False))

    # ---------------------- 4. GENERATE GRAPHS ----------------------
    chart_specs = [
        ("Throughput_Gbps", "Throughput (Gbps)",
         "Throughput by User Load", "throughput_comparison.png"),
        ("Utilization_Percent", "Capacity utilization (%)",
         "Capacity Utilization by User Load", "utilization_comparison.png"),
        ("Estimated_Delay_ms", "Estimated delay (ms)",
         "Illustrative Delay by User Load", "delay_comparison.png"),
```

```
    ("Estimated_Delay_ms", "Estimated delay (ms)",
     "Illustrative Delay by User Load", "delay_comparison.png"),
    ("Handoffs_Per_Hour", "Estimated handoffs per hour",
     "Handoff Activity by User Load", "handoff_comparison.png"),
]

for metric, y_label, title, filename in chart_specs:
    plt.figure(figsize=(9, 5.5))
    for strategy in STRATEGIES:
        subset = df[df["Strategy"] == strategy]
        plt.plot(
            subset["Users"], subset[metric],
            marker="o", linewidth=2, label=strategy
        )
    plt.xlabel("Number of active users")
    plt.ylabel(y_label)
    plt.title(title)
    plt.xticks(USER_LOADS, [f"{x:,}" for x in USER_LOADS])
    plt.grid(True, linestyle="--", alpha=0.5)
    plt.legend()
    plt.tight_layout()
    plt.savefig(OUTPUT_DIR / filename, dpi=200)
    plt.close()

print(f"\nDone. Tables and graphs are saved in: {OUTPUT_DIR.resolve()}")
print("Note: handoff counts are assumed-rate estimates, not measured handover events.")
```

Appendix B – Important interpretation notes

- All capacities, traffic rates, cell radii, handover rates and spectrum values are planning assumptions because the assignment brief does not provide real operator or site-survey data.
- The geometric cell-area calculation does not account for overlap, terrain, indoor loss or antenna patterns.
- The access-attempt estimate is not a detailed PRACH configuration calculation.
- The delay curve is illustrative and must not be presented as a measured 5G latency result.
- The handover totals are estimated events based on assumed rates, not observed handover success or failure.
- Before real deployment, perform a radio survey, spectrum and regulatory review, backhaul assessment, detailed RAN simulation and field acceptance testing.